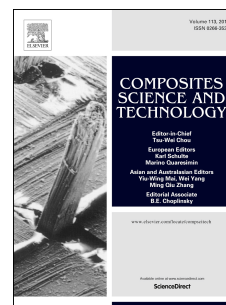


Accepted Manuscript

Suppression of elevated temperatures space charge accumulation in polypropylene/elastomer blends by deep traps induced by surface-modified ZnO nanoparticles

Bin Dang, Qi Li, Yao Zhou, Jun Hu, Jinliang He



PII: S0266-3538(17)31653-6

DOI: [10.1016/j.compscitech.2017.10.005](https://doi.org/10.1016/j.compscitech.2017.10.005)

Reference: CSTE 6930

To appear in: *Composites Science and Technology*

Received Date: 15 July 2017

Revised Date: 13 September 2017

Accepted Date: 5 October 2017

Please cite this article as: Dang B, Li Q, Zhou Y, Hu J, He J, Suppression of elevated temperatures space charge accumulation in polypropylene/elastomer blends by deep traps induced by surface-modified ZnO nanoparticles, *Composites Science and Technology* (2017), doi: 10.1016/j.compscitech.2017.10.005.

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Suppression of elevated temperatures space charge accumulation in polypropylene/elastomer blends by deep traps induced by surface-modified ZnO nanoparticles

Bin Dang, Qi Li*, Yao Zhou, Jun Hu and Jinliang He

The State Key Lab of Power System, Department of Electrical Engineering, Tsinghua University, Beijing 100084, China

ABSTRACT

Space charge accumulation is a critical issue for the deterioration of the elevated temperature insulating property of polymeric materials. We present the influence of surface-modified ZnO nanoparticles on the space charge distribution and direct current (DC) resistivity of polypropylene (PP)/elastomer blends under elevated temperature. Octyltrimethoxysilane, a silane coupling agent, was used for the surface modification of nanoparticles. Morphology characterization results indicated that the elastomer and coated ZnO were well dispersed in the PP matrix. It was observed that the coated ZnO can significantly improve the insulating properties, including a minimized electric field distortion (4.0%) and increased DC volume resistivity ($1.41 \times 10^{18} \Omega \cdot m$) under an electric field of 40kV/mm and 70°C. For the same reason, the DC resistivity of 2 phr PP/elastomer/ZnO ternary composites was improved by 13.4 times compared with that of pure PP/elastomer blends. The suppression of space charges may originate from deep traps existing in spherulite boundaries and interfacial zones between polypropylene and ZnO. This work provides an effective method to endow a recyclable insulating material with outstanding elevated temperature insulating performances.

KEYWORDS: Surface treatments; Nano composites; Space-charge Electrical properties; High-temperature properties; Interface.

*Corresponding author. Tel: +86-010-62788811. E-mail: qili1020@tsinghua.edu.cn (Qi Li)

With its excellent electrical insulating performance, crosslinking polyethylene (XLPE) and its nanocomposites are considered as the most important dielectric materials for commercial high voltage direct current (HVDC) cable insulation [1-6]. However, owing to the thermosetting nature of XLPE, it has to be processed through burring or burning when the cables reach the end of their service lifetime, which causes a large adverse impact on the environment [7-9]. A large number of studies have been carried out to explore environmentally friendly materials for HVDC cables, for example, high density polyethylene (HDPE)/low density polyethylene (LDPE) blends [10, 11]; and isotactic polypropylene (iPP)/elastomer blends [12-16]. Among these polymer blends, iPP/elastomer blends, which have even better insulating properties than HDPE/LDPE blends, are promising materials for large-capacity HVDC power cables[16]. Meanwhile, it has been found that iPP/elastomer blends as cable insulation also have a higher operating temperature than HDPE/LDPE blends, which is another key asset for large-capacity HVDC [17].

Nevertheless, it has generally been observed that the significant space charge accumulation in iPP/elastomer blends [16] can result in the distortion of the electric field and exert a deteriorative effect on the long-term operation and reliability of insulation materials [18, 19]. In this context, great effort has been made to suppress the space charge through the addition of a small amount of nanoparticles (NPs), including semiconducting (ZnO [20, 21] and TiO₂ [22, 23]) and insulating particles (MgO [24, 25], SiO₂ [26, 27]), because polymer matrices incorporated with these nanoparticles show optimized space charge distribution. Among these materials, ZnO is unique by virtue of the outstanding ability to resist UV damage to the power cables [28-30].

However, owing to the Joule heating by the conductor through which the intensive current flows, the temperature of the cable insulation can generally reach up to 70 °C, and this value increases with increasing the transmission capacity, resulting in a largely reduced charge injection barrier and greatly increased charge

mobility [27]. These variations may cause a more severe space charge accumulation than that at room temperature. Nevertheless, studies on elevated temperature space charge dynamics in iPP/elastomer blends are rare [31], because the characterization of the space charge at elevated temperature is challenging.

In our research, PP/elastomer blends incorporated with ZnO NPs modified by a silane coupling agent are selected as the material platform. The space charge distribution and conduction experiment of PP/elastomer/ZnO nanocomposites were conducted with temperature as the primary factor to be taken into account. The optimal content of the surface-modified ZnO NPs on the electrical properties of the PP/elastomer matrix was determined experimentally. The relationship between the phase structure and the space charge accumulation in the PP/elastomer nanocomposites at elevated temperature is discussed for the first time. ~~The experimental data suggest that the presence of the surface functionalized nanoparticles can remarkably reduce the space charge accumulation and improve the electrical resistivity under high temperature.~~ The mechanism of the space charge suppression at elevated temperature in these nanocomposites is analyzed and discussed. The results of this study may have an influence on the future development and engineering of recyclable HVDC cable insulations, especially those that may experience harsh temperature conditions.

2. Material and methods

2.1. Materials

PP with a melt flow index of 4 g/10 min, density of 0.9 g/cm³ was used as the base material. Propylene-based elastomers, namely propylene alpha-olefin copolymer (PO), was used to improve the mechanical properties of PP. Octyltrimethoxysilane (C8) (Heowns Biochem Technologies) was used to modify the surface properties of ZnO NPs (Sigma-Aldrich). An average diameter of NPs was approximately 100 nm.

2.2. Surface modification of the ZnO NPs

Anhydrous surface modification of ZnO NPs was carried out in n-heptane to form a uniform surface coverage. ZnO powder (5g) was dispersed in n-heptane (300 ml) by ultrasonic agitation and transferred to an Erlenmeyer flask. C8 (5mL) was then added to the ZnO/heptane solution under rapid magnetic stirring (500 rpm) for 24 hours at room temperature, after which the NPs were washed three times with ethanol to eliminate the extra silanes by using a centrifugal machine and ultrasonic bath. The modified NPs will herein be referred to as ZnO-C8.

2.3. Preparation of the PP/PO/ZnO nanocomposites

Table 1 shows the ternary polymer composites containing the rigid fillers and soft elastomer used in this investigation. The nanocomposites were prepared as follows. The adequate quality ZnO-C8 powder was first mixed uniformly with PO at 160 °C for 10 minutes; then the PO/ZnO-C8 was mixed with PP for 10 minutes at 175 °C. To prepare the test samples for the space charge and conduction measurements, the mixture was premelted at 200 °C for 10 minutes, molded under at a pressure of approximately 20 MPa for 5 minutes and cooled to 25 °C with a pressure of approximately 10 MPa for another 5 minutes.

Table 1. Materials used in this investigation.

Designation	PP (wt %)	PO (wt %)	ZnO-C8 (wt %)
0 phr	60	40	0
0.5 phr	60	40	0.5
1 phr	60	40	1
2phr	60	40	2
3 phr	60	40	3

2.4. Instrumentation and characterization methods

Fourier transform infrared spectroscopy (FTIR) between 400 and 4000 cm^{-1} was recorded on a Thermo Scientific Nicolet iS10 spectrophotomete. A given amount ($3 \pm 0.5\text{mg}$) of NPs was mixed with a suitable amount of KBr (300mg) and then the mixtures were pressed for 60 s. The number of scans was 32.

The melting and crystallization of the nanocomposites with a C8 coupling agent were investigated by TA instruments differential scanning calorimetry (DSC) 2000. The sample (5 ± 0.5 mg) was first heated from room temperature to 200 °C at a rate of 10 °C/min, then maintained for 5 minutes to eliminate the heat thermal history and finally cooled to 20 °C at a rate of 10 °C/min to measure the crystallization. The melting peak was recorded from the second heating from 20 to 200 °C. The degree of crystallinity (X_c) of PP was calculated by the melting enthalpies according to:

$$X_c = \Delta H_m / \Delta H_m^\infty \quad (1)$$

where ΔH_m represents the measured melting enthalpy of the blends, ΔH_m^∞ refers to the melting enthalpy for 100% crystalline polypropylene (209 J/g) [32, 33].

A Hitachi SU8010 field emission scanning electron microscope (FE-SEM) was used to obtain the dispersion and distribution of the elastomer and NPs in polypropylene. The 600 μm thick nanocomposites were brittle fractured in liquid nitrogen, and then the fracture surface was coated with platinum for 60 s at a 20 mA operating current in the magnetron sputtering equipment.

2.5. Space charge measurement

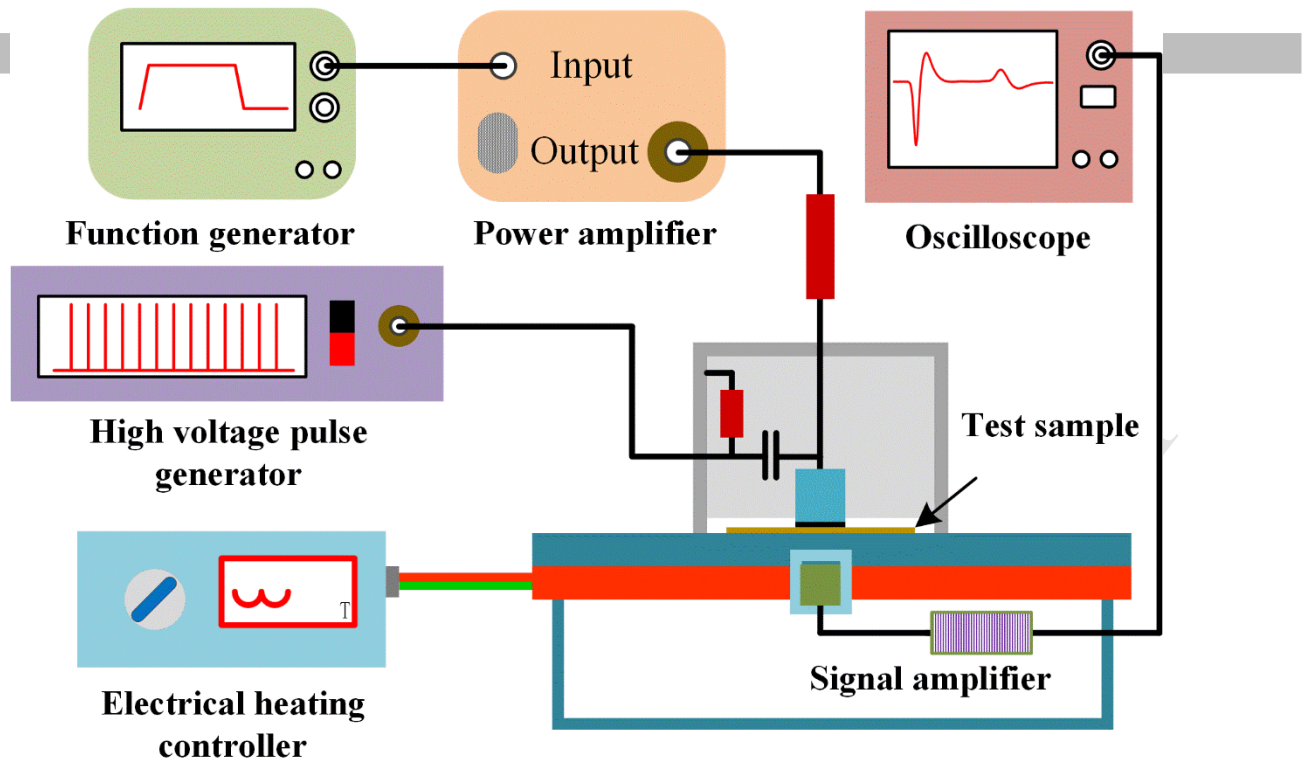


Fig. 1 Diagram of the improved space charge measurement system under high temperature.

An improved pulsed electro-acoustic method (PEA) system with a heated low electrode has been developed to measure the space charge at a temperature higher than 100 °C presented in Figure 1. A new P(VDF-TrFE) film, whose Curie temperature is above 120 °C, has been adopted as the piezoelectric transducer. The details of the basic PEA measurement are described elsewhere [34-36]. The measurement experiments were conducted at 70 °C with semiconductor electrodes. The space charge distribution was measured for 10 minutes under a 40 kV/mm electrical field and 800V pulse voltage with a duration of 5 ns. Considering that we only focus on the results with relative values in this research, no attenuation recovery was obtained.

2.6. High field electrical conduction measurement

Charging current and DC volume resistivity measurement experiments under high temperature were conducted using the three-terminal-cell system illustrated in Figure 2. To eliminate the internal charges, the sample was first electrically short-circuited for 30 min. Then the charging current was performed for 30 min under an electric field of 40kV/mm and a temperature of 70 °C.

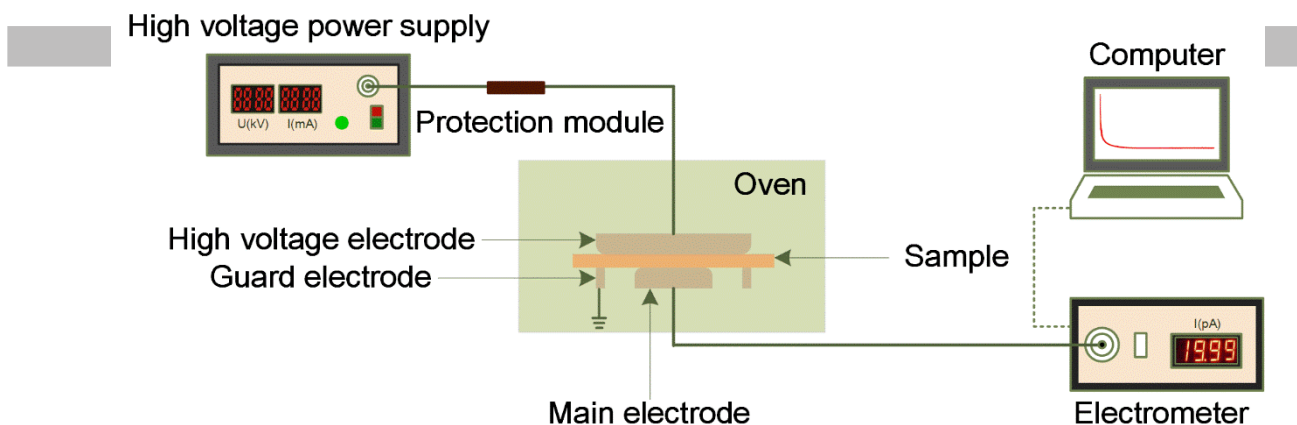


Fig. 2 Diagram of the leakage current measurement under high temperature.

3. Results and Discussion

3.1. Characterization of the surface-modified ZnO NPs

Figure 3 presents the FTIR results of the pristine and silane-coated NPs. The absorbance bands at 450–600 cm^{-1} originated from ZnO stretching. Compared with the spectrum of the pristine NPs, a new triple peak of C-H stretching bands at 2960 cm^{-1} , 2924 cm^{-1} , 2854 cm^{-1} appeared for ZnO-C8 NPs, indicating that the pristine NPs were successfully coated by C8. The hydroxyl (O-H) bending vibration band at 3442 cm^{-1} in the spectra of C8-coated NPs was lower than that of the pristine NPs, suggesting that some surface O-H groups participated in the grafting reaction.

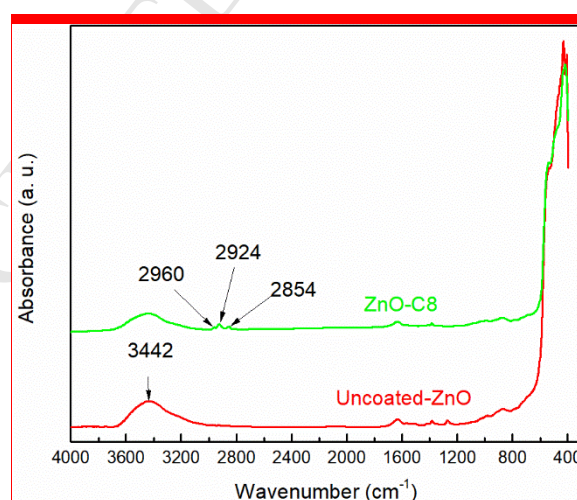


Fig. 3 FTIR spectra of uncoated ZnO and C8–modified ZnO.

3.2. Dispersion and morphology

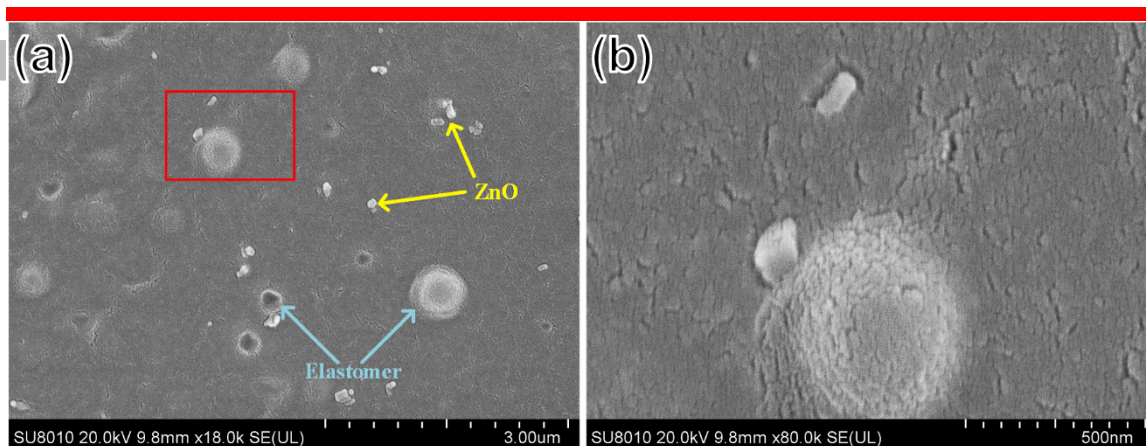


Fig. 4 Scanning electron micrograph showing 3 phr ZnO-C8 in PP/PO blends; (b) is a partial enlargement of the frame shown in (a).

Figure 4 shows the phase structure acquired for the PP/PO/ZnO-C8 blends. From the SEM image, the bigger particles were attributed to the elastomer PO and the smaller particles represented ZnO-C8 particles. Considering that the ZnO-C8 was mixed with PO in advance, it is interesting to note that the isolated ZnO-C8 NPs only appeared in the PP matrix. The phase structure of the present ternary composites depends on both kinetic parameters, which are controlled by the viscosity ratio between the matrix and elastomer, and thermodynamic control, which is dominated by the matrix-filler and elastomer-filler interactions [37]. The size of ZnO-C8 NPs was less than 200 nm in the PP matrix, indicating outstanding the dispersibility of ZnO-C8 in PP. The matrix and elastomer exhibited a typical sea-island structure. With respect to their sizes, these elastomer phases were quite small (less than 500 nm), suggesting an excellent compatibility of the matrix and elastomer.

3.3. Crystallization characteristics

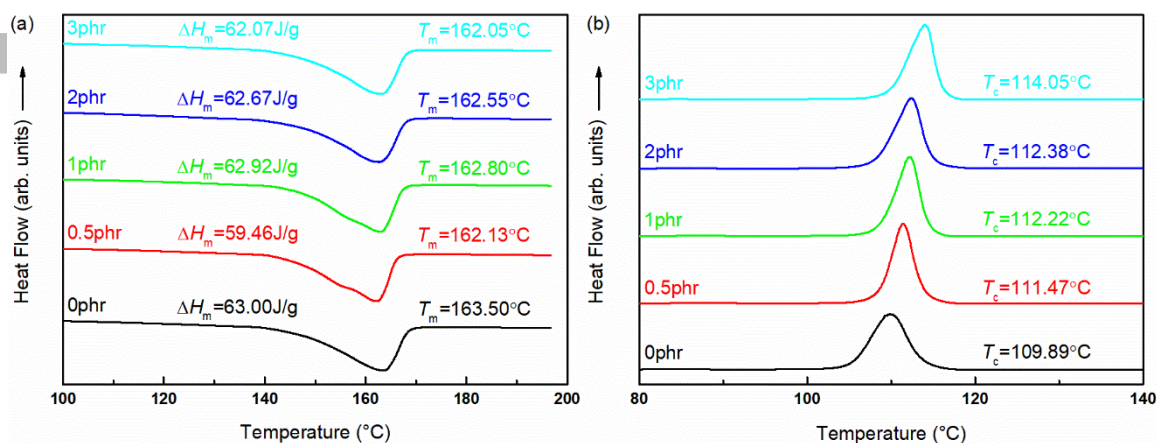


Fig. 5 (a) Melting and (b) non-isothermal crystallization curves of the PP/PO blends and their nanocomposites.

Figure 5 shows the crystallization and melting behaviors of the PP/PO blends and their nanocomposites. Despite no significant changes in the melting peak temperature (162–164 °C) and crystallinity (28–30%) were observed when NPs were added to the PP/PO blends, the crystallization temperature remarkably increased with an increasing amount (0.5 to 3 phr) of surface-modified ZnO NPs. This suggested that the ZnO-C8 NPs had a prominent nucleating effect on the polymer matrix, which led to larger interfacial areas in the spherulite boundaries for nanocomposites than that for the pure PP/PO blends.

3.4. Space charge properties

The accumulation of space charge has a great influence on both the short term and long term DC electrical performance of polymer dielectrics[38]. The space charge distributions during polarization in five samples are shown in Figure 6(a). Figure 6(a1) presents the charge injection and transport versus time in the neat PP/PO blends. A large homo space charge started to inject at both electrodes and migrated across the sample. With an increase of time, the positive charge and negative charge recombined at the center of the sample. The space charge tended to be stable at 10 mins of voltage application. Figure 6(a2) shows that only a little charge accumulated around the electrode and a few recombinations occurred in the nanocomposites containing 0.5 phr ZnO-C8 NPs. Moreover, it can be seen from Figure 6(a3), 6(a4) and 6 (a5) that with the increasing amount of coated ZnO NPs, only a very small amount of homo space charge was observed in the

vicinity of the electrodes and few recombinations appeared in the entire samples.

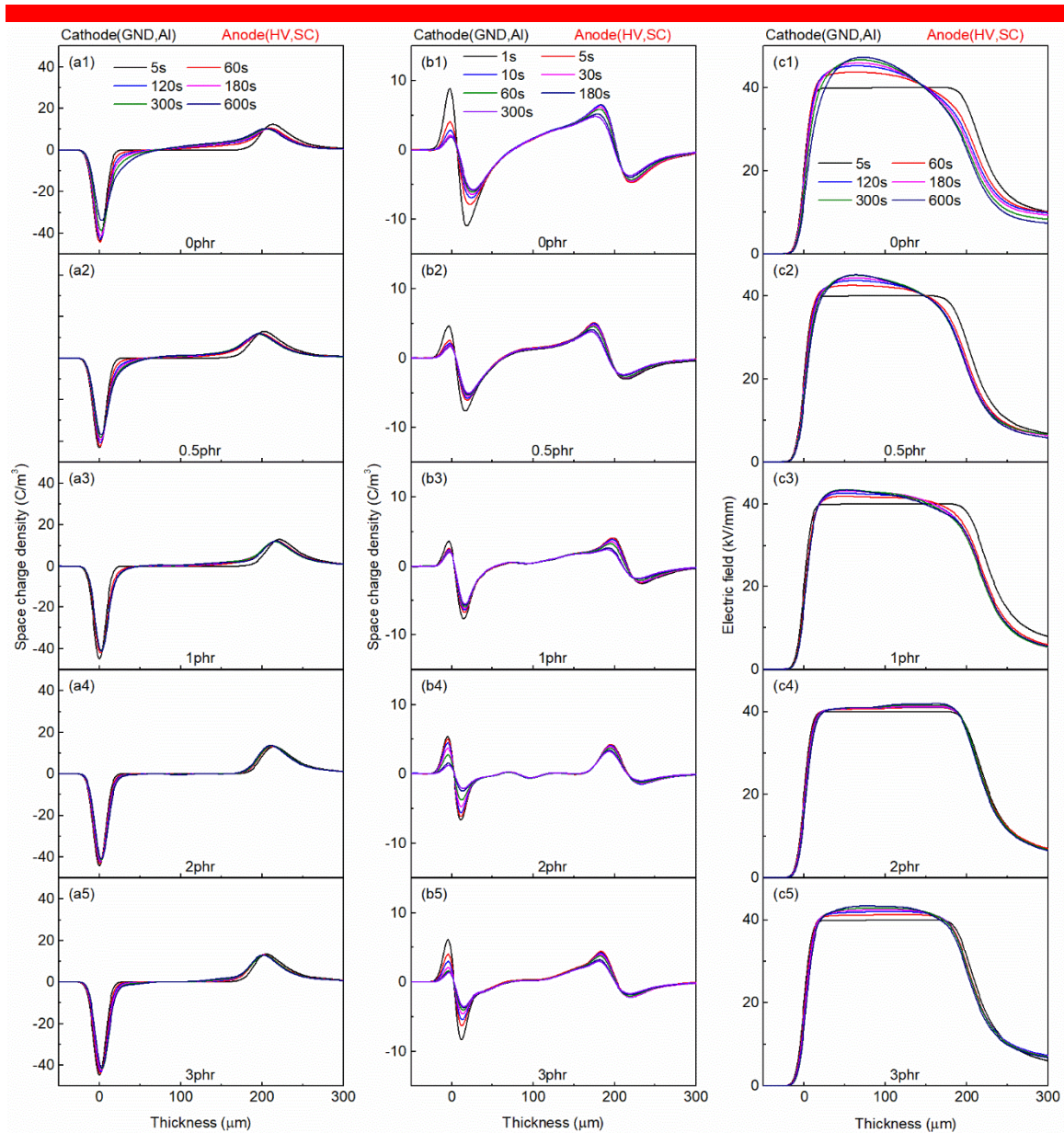


Fig. 6 (a) Space charge accumulation during polarization, (b) space charge decay during depolarization and (c) electric field distribution during polarization of PP/PO blends and their nanocomposites from 0 to 3 phr of ZnO-C8 loading, subjected to an applied electric field of 40 kV/mm at 70 °C.

To directly evaluate the amount of charge accumulation in the PP/PO blends and their nanocomposites, the space charge distribution with voltage-off was measured and is shown in Figure 6(b). It can be visually seen from Figure 6(b1) that after a sudden short circuit, a large amount of positive charge and negative

charge rapidly accumulated near the anode and cathode, respectively. The homocharge dissipated at a fast rate with the increase of time. Figure 6(b2), (b3), (b4) and (b5) confirm the significant suppression of space charge in nanocomposites. Among these nanocomposites, 2 phr nanocomposites exhibited the best ability for space charge inhibition.

The electric field distortion will severely hinder to insulation materials under elevated temperature [18, 39]. The electric field distribution during polarization of nanocomposites and the pristine PP/PO blends is presented in Figure 6(c). All sample showed an increasing electric field distortion with an increase in time. The minimum electric field distortion was found for composites with 2 phr ZnO-C8 NPs.

3.5. Quantitative results of space charge characteristics

To accurately analyze the influence of the coated ZnO NPs to PP/PO blends, it is essential to obtain the quantitative results of the total injected charge during voltage-on, maximum electric field during voltage-on and total dissipated charge during voltage-off as function of time. The total amount of the injected charge in the sample during polarization $q_p(t)$ and the total amount of charge in the sample during depolarization $q_d(t)$ are acquired from the space charge density according to:

$$q_p(t) = \int_{x_1}^{x_2} |\rho(x, t)| S dx - \int_{x_1}^{x_2} |\rho(x, t_0)| S dx \quad (2)$$

$$q_d(t) = \int_{x_1}^{x_2} |\rho(x, t)| S dx \quad (3)$$

where the first item of Equation (2) refers to the total charge in the sample including injected charge and capacitive charge, and the latter item of Equation (2) refers to the capacitive charge. Specifically, ρ is the space charge density, t is the time coordinate, x is the spatial coordinate, S is the piezoelectric transducer area, x_1 is the start position of the cathode peak, x_2 is the end position of the anode peak and t_0 is the start time of the charge injection. The electric field $E(x, t)$, the maximum electric field $E_{\max}(t)$ and the distortion factor of electric field $k(t)$ can be obtained as follows:

$$E(x,t) = \frac{1}{\epsilon_0 \epsilon_r} \int_{x1}^x |\rho(x,t)| S dx \quad \text{ACCEPTED MANUSCRIPT} \quad (4)$$

$$E_{\max}(t) = \max_x E(x,t) \quad (5)$$

$$k(t) = \frac{E_{\max}(t) - E_p(t)}{E_p(t)} \times 100\% \quad (6)$$

where ϵ_0 is the permittivity of vacuum, ϵ_r is the relative permittivity and $E_p(t)$ is the Laplace field.

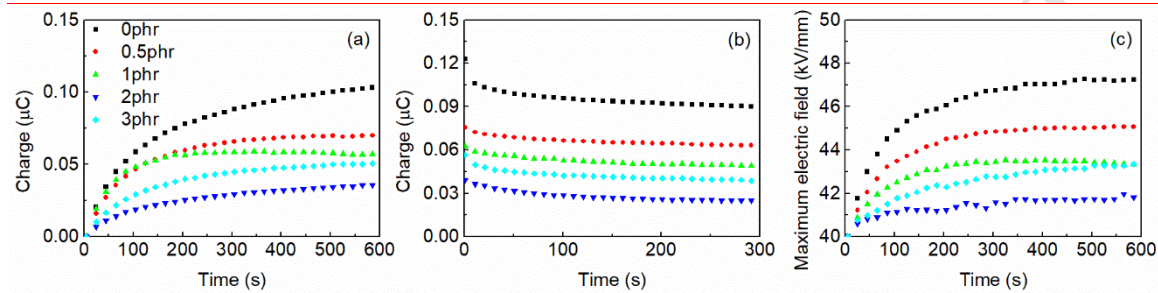


Fig. 7 (a) Total space charge during polarization, (b) total space charge during depolarization and (c) maximum electric field during polarization of PP/PO/ZnO nanocomposites from 0 to 3 phr of ZnO-C8 loading, subjected to an applied electric field of 40 kV/mm at 70 °C.

Table 3. Summary of PEA results of PP/PO blends and their nanocomposites based on different weight fractions of ZnO NPs at $E=40\text{kV/mm}$ and $T=70\text{ °C}$.

	0 phr	0.5 phr	1 phr	2phr	3 phr
Space charges q_p (600) (μC)	0.1069	0.0737	0.0609	0.0385	0.0515
Space charges q_d (900) (μC)	0.0899	0.0627	0.0489	0.0249	0.0391
Maximum electrical strength $E_{\max}$ (600) (kV/mm)	47.27	45.03	43.34	41.61	43.30
Distortion factor k (600) (%)	18.2	12.6	8.4	4.0	8.3

Figure 7(a) and (c) show the time dependence of the total space charge and the maximum electric field during polarization, respectively, in PP/PO blends and their nanocomposites under 40 kV/mm and 70 °C. Both the total space charge and maximum electric field were lower during the entire polarization for all composites compare with those of the unfilled PP/PO blends. Both the total injected charge and maximum

electric field during voltage-on decreased with an increasing amount (0.5 to 2 phr) of surface-modified ZnO NPs. The total space charge and maximum electric field of 2 phr nanocomposites maintained the lowest values among all measurement times. The total space charge during depolarization is illustrated in Figure 7(b). It can be found that the total charge rapidly dissipated and slowly decayed as follows.

The values of q_p (600), q_d (900), $E_{\max}$ (600) and k (600) are listed in Table 3. The addition of 2 phr C8-coated ZnO NPs show minimum q_p (600), q_d (900), and k (600).

3.6. High field electrical conduction

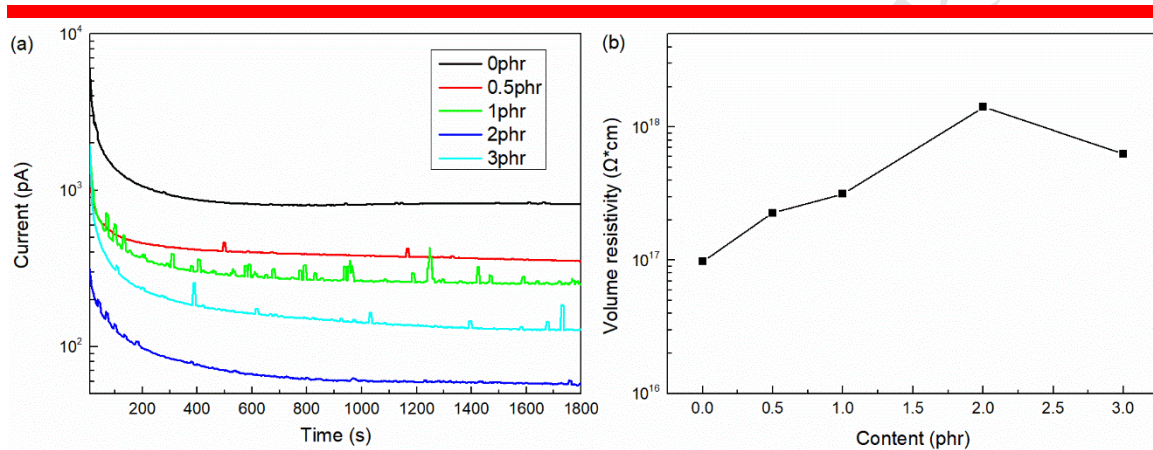


Fig. 8 (a) Charging current and (b) volume resistivity of PP/PO blends and their nanocomposites subjected to an applied electric field of 40 kV/mm at 70 °C.

Steady DC charging current, which determines DC losses, has a great effect on the insulation material of HVDC cables. A higher DC volume resistivity results in lower DC losses, which causes a minimized temperature rise in the dielectric and thus improves its service lifetime [40-42]. Figure 8a shows the charging current of PP/PO blends and their nanocomposites based on coated NPs (0.5–3 phr) as a function of time. The dashed black line shows the damped charging current of the unfilled PP/PO blends. A small drop in charging current can be observed for nanocomposites with 0.5 phr ZnO-C8 NPs. The nanocomposites with 2 phr ZnO-C8 NPs had the lowest charging current compared with the other samples during the measurement process.

Figure 8b presents the volume resistivity values based on measured leakage current after 30min as a function of ZnO-C8 content. For the PP/PO blends filled with the as-received ZnO NPs coated by C8 silane, the volume resistivity values were calculated to be 2.27×10^{17} , 3.15×10^{17} , 1.41×10^{18} and $6.28 \times 10^{17} \Omega \cdot \text{cm}$, for the 0.5, 1, 2 and 3 phr samples, respectively. It was clear that with the addition of 2 phr surface-modified ZnO NPs, the volume resistivity was increased by 13.4 times.

3.7. Mechanism for space charge suppression

It is very difficult to fully understand the mechanism for space charge suppression in insulating materials where both electronic conduction and ionic conduction contribute to the leakage current. However, it can be learned from space charge dynamics during polarization that current conduction in PP/PO blends and their nanocomposites mainly originated from electronic conduction, owing to the high purity of the samples. Because of the lowered Schottky barrier under an elevated temperature, the carriers can be easily injected and accumulate in the pure PP/PO blends. The carrier trap sites play a dominant role in improving the space charge distribution in polymer nanocomposites [27, 43-47]. The reduced conduction current under elevated temperature in nanocomposites demonstrated that large numbers of deep traps have been induced by ZnO NPs. The deep traps in PP/PO blends doped by ZnO-C8 NPs can strongly capture the charge injected from the electrodes, and the trapped charge near the electrodes in turn increases the barrier against charge injection. Therefore, the space charge in PP/PO nanocomposites is significantly suppressed.

The deep traps are generally considered to be induced by interfaces in polymers. Growing interfacial regions, including spherulite boundaries (I) and interfacial regions between PP and ZnO-C8 (II), were observed in the PP/PO nanocomposites shown in Figure 9. It has been proved that PP with smaller spherulite show fewer space charges [48], indicating that a number of deep traps exist at the spherulite boundaries. Additionally, owing to the noteworthy structural disorder in interfacial regions (II), there are numerous defects acting as deep traps [44, 49, 50]. Hence, both interfacial regions (I) and (II) have a large contribution

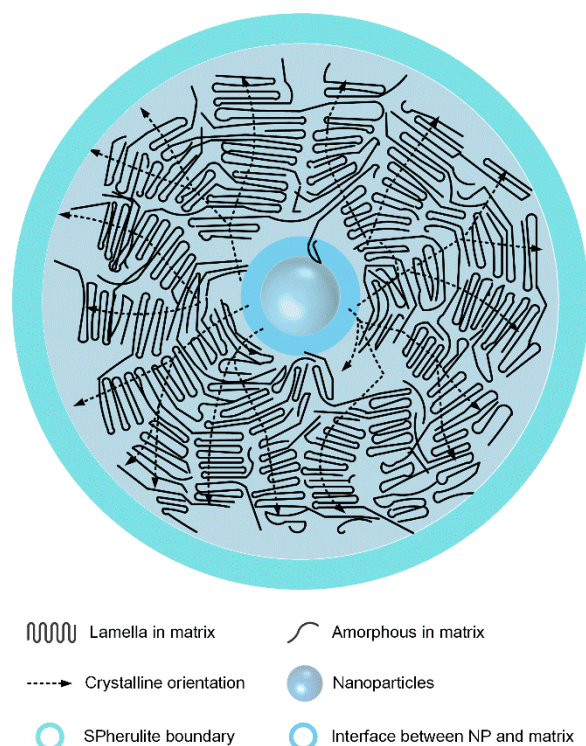


Fig. 9 Increased interfacial regions in PP/PO/ZnO-C8 nanocomposites.

4. Conclusions

Electrical insulating materials based on PP/PO blends and ZnO NPs (≤ 3 phr) coated by C8 silane have been fabricated for potential applications as dielectric materials in recyclable HVDC cables. The effect of ZnO-C8 on the electrical insulating properties of PP/PO blends were investigated by measuring the space charge distribution and charging current under high electric field and elevated temperature. Among these nanocomposites, PP/PO blends filled with 2 phr ZnO-C8 exhibited the best DC electrical insulation characteristics, namely the least space charge accumulation and highest volume resistivity. The space charge was suppressed by deep traps may originating from spherulite boundaries and interfaces between PP and ZnO-C8. This work provides an understanding of the origin for the excellent elevated temperature electrical insulating properties of polymer dielectrics doped with surface-modified nanoparticles.

This work was supported in part by the National Basic Research Program of China (973 Project) under grant 2014CB239504. We thank Liwen Bianji, Edanz Group China (www.liwenbianji.cn/ac), for editing the English text of a draft of this manuscript.

References

- [1] G. Mazzanti, M. Marzotto, Extruded cables for high-voltage direct-current transmission: advances in research and development, John Wiley & Sons, Hoboken, 2013.
- [2] N. Hozumi, T. Takeda, H. Suzuki, T. Okamoto, Space charge behavior in XLPE cable insulation under 0.2-1.2 MV/cm dc fields, *IEEE Transactions on Dielectrics and Electrical Insulation* 5(1) (1998) 82-90.
- [3] N. Hozumi, H. Suzuki, T. Okamoto, K. Watanabe, A. Watanabe, Direct observation of time-dependent space charge profiles in XLPE cable under high electric fields, *IEEE Transactions on Dielectrics and Electrical Insulation* 1(6) (1994) 1068-1076.
- [4] J. Fothergill, G. Montanari, G. Stevens, C. Laurent, G. Teyssedre, L. Dissado, U. Nilsson, G. Platbrood, Electrical, microstructural, physical and chemical characterization of HV XLPE cable peelings for an electrical aging diagnostic data base, *IEEE Transactions on Dielectrics and Electrical Insulation* 10(3) (2003) 514-527.
- [5] X. Zheng, G. Chen, Propagation mechanism of electrical tree in XLPE cable insulation by investigating a double electrical tree structure, *IEEE Transactions on Dielectrics and Electrical Insulation* 15(3) (2008).
- [6] R. Densley, An investigation into the growth of electrical trees in XLPE cable insulation, *IEEE transactions on electrical insulation* (3) (1979) 148-158.
- [7] J. Zha, H. Yan, W. Li, Z. Dang, Morphology and crystalline-phase-dependent electrical insulating properties in tailored polypropylene for HVDC cables, *Applied Physics Letters* 109(22) (2016) 222902.
- [8] K. Yoshino, T. Demura, M. Kawahigashi, Y. Miyashita, K. Kurahashi, Y. Matsuda, Application of a novel polypropylene to the

- [9] K. Kurahashi, Y. Matsuda, Y. Miyashita, T. Demura, A. Ueda, K. Yoshino, The application of novel polypropylene to the insulation of electric power cable (3), *Electrical Engineering in Japan* 155(3) (2006) 1-8.
- [10] Y. Lin, W. Du, D. Tu, W. Zhong, Q. Du, Space charge distribution and crystalline structure in low density polyethylene (LDPE) blended with high density polyethylene (HDPE), *Polymer international* 54(2) (2005) 465-470.
- [11] C. Green, A. Vaughan, G. Stevens, S. Sutton, Recyclable power cable comprising a blend of slow-crystallized polyethylenes, *IEEE Transactions on Dielectrics and Electrical Insulation* 20(1) (2013) 1-9.
- [12] Y. Zhou, J. He, J. Hu, X. Huang, P. Jiang, Evaluation of polypropylene/polyolefin elastomer blends for potential recyclable HVDC cable insulation applications, *IEEE Transactions on Dielectrics and Electrical Insulation* 22(2) (2015) 673-681.
- [13] B. Dang, J. He, J. Hu, Y. Zhou, Large improvement in trap level and space charge distribution of polypropylene by enhancing the crystalline/amorphous interfaces effect in blends, *Polymer international* 65(4) (2016) 371-379.
- [14] C. Green, A. Vaughan, G. Stevens, A. Pye, S. Sutton, T. Geussens, M. Fairhurst, Thermoplastic cable insulation comprising a blend of isotactic polypropylene and a propylene-ethylene copolymer, *IEEE Transactions on Dielectrics and Electrical Insulation* 22(2) (2015) 639-648.
- [15] G. Montanari, D. Fabiani, F. Palmieri, D. Kaempfer, R. Thomann, R. Mulhaupt, Modification of electrical properties and performance of EVA and PP insulation through nanostructure by organophilic silicates, *IEEE Transactions on Dielectrics and Electrical Insulation* 11(5) (2004) 754-762.
- [16] J. Diao, X. Huang, Q. Jia, F. Liu, P. Jiang, Thermoplastic isotactic polypropylene/ethylene-octene polyolefin copolymer nanocomposite for recyclable HVDC cable insulation, *IEEE Transactions on Dielectrics and Electrical Insulation* 24(3) (2017) 1416-1429.
- [17] B. Dang, Y. Zhou, J. He, J. Hu, Relationship between space charge behaviors and trap level distribution in polypropylene/propylene ethylene rubber/ZnO nanocomposites, *Electrical Insulation and Dielectric Phenomena (CEIDP)*, 2016 IEEE Conference on, IEEE, 2016, pp. 595-598.

- [18] Y. Zhang, J. Lewiner, C. Alquie, N. Hampton, Evidence of strong correlation between space-charge buildup and breakdown in cable insulation, *IEEE Transactions on Dielectrics and Electrical Insulation* 3(6) (1996) 778-783.
- [19] C. Laurent, G. Teyssedre, S. Le Roy, F. Baudoin, Charge dynamics and its energetic features in polymeric materials, *IEEE Transactions on Dielectrics and Electrical Insulation* 20(2) (2013) 357-381.
- [20] B. Du, H. Xu, J. Li, Z. Li, Space charge behaviors of PP/POE/ZnO nanocomposites for HVDC cables, *IEEE Transactions on Dielectrics and Electrical Insulation* 23(5) (2016) 3165-3174.
- [21] F. Tian, J. Yao, P. Li, Y. Wang, M. Wu, Q. Lei, Stepwise electric field induced charging current and its correlation with space charge formation in LDPE/ZnO nanocomposite, *IEEE Transactions on Dielectrics and Electrical Insulation* 22(2) (2015) 1232-1239.
- [22] Y. Zhou, J. Hu, B. Dang, J. He, Titanium oxide nanoparticle increases shallow traps to suppress space charge accumulation in polypropylene dielectrics, *RSC Advances* 6 (2016) 48720-48727.
- [23] Y. Zhou, J. Hu, B. Dang, J. He, Effect of different nanoparticles on tuning electrical properties of polypropylene nanocomposites, *IEEE Transactions on Dielectrics and Electrical Insulation* 24(3) (2017) 1380-1389.
- [24] Y. Zhou, J. He, J. Hu, B. Dang, Surface-modified MgO nanoparticle enhances the mechanical and direct-current electrical characteristics of polypropylene/polyolefin elastomer nanodielectrics, *Journal of Applied Polymer Science* 133(1) (2016).
- [25] Z. Li, W. Cao, G. Sheng, X. Jiang, M. Danikas, Experimental study on space charge and electrical strength of MgO nano-particles/polypropylene composite, *IEEE Transactions on Dielectrics and Electrical Insulation* 23(3) (2016) 1812-1819.
- [26] B. Dang, J. He, J. Hu, Y. Zhou, Tailored sPP/Silica Nanocomposite for Ecofriendly Insulation of Extruded HVDC Cable, *Journal of Nanomaterials* 2015 (2015) 686248.
- [27] L. Zhang, M. Khani, T. Krentz, Y. Huang, Y. Zhou, B. Benicewicz, J. Nelson, L. Schadler, Suppression of space charge in crosslinked polyethylene filled with poly (stearyl methacrylate)-grafted SiO₂ nanoparticles, *Applied Physics Letters* 110(13) (2017) 132903.
- [28] H. Zhao, R. Li, A study on the photo-degradation of zinc oxide (ZnO) filled polypropylene nanocomposites, *Polymer* 47(9)

- [29] L. Brambilla, G. Consolati, R. Gallo, F. Quasso, F. Severini, Environmental degradation of isotactic polypropylene plates as studied by positron annihilation lifetime spectroscopy, *Polymer* 44(4) (2003) 1041-1044.
- [30] S. Chandramouleeswaran, S. Mhaske, A. Kathe, P. Varadarajan, V. Prasad, N. Vigneshwaran, Functional behaviour of polypropylene/ZnO-soluble starch nanocomposites, *Nanotechnology* 18(38) (2007) 385702.
- [31] Y. Zhou, J. Hu, J. He, Space charge behavior in polypropylene/polyolefin elastomer/MgO nanocomposites under temperature gradient, *Electrical Insulation and Dielectric Phenomena (CEIDP)*, 2016 IEEE Conference on, IEEE, 2016, pp. 599-602.
- [32] D. SAL, M. Tavares, D. Politano, F. Coutinho, M. Rocha, Polymer blends based on polyolefin elastomer and polypropylene, *Journal of Applied Polymer Science* 66(10) (1997) 2005-2014.
- [33] T. McNally, P. McShane, G. Nally, W. Murphy, M. Cook, A. Miller, Rheology, phase morphology, mechanical, impact and thermal properties of polypropylene/metallocene catalysed ethylene 1-octene copolymer blends, *Polymer* 43(13) (2002) 3785-3793.
- [34] N. Ahmed, N. Srinivas, Review of space charge measurements in dielectrics, *IEEE Transactions on Dielectrics and Electrical Insulation* 4(5) (1997) 644-656.
- [35] Y. Li, M. Yasuda, T. Takada, Pulsed electroacoustic method for measurement of charge accumulation in solid dielectrics, *IEEE Transactions on Dielectrics and Electrical Insulation* 1(2) (1994) 188-195.
- [36] T. Maeno, K. Fukunaga, High-resolution PEA charge distribution measurement system, *IEEE Transactions on Dielectrics and Electrical Insulation* 3(6) (1996) 754-757.
- [37] C. Ma, Y. Mai, M. Rong, W. Ruan, M. Zhang, Phase structure and mechanical properties of ternary polypropylene/elastomer/nano-CaCO₃ composites, *Composites Science and Technology* 67(14) (2007) 2997-3005.
- [38] G. Montanari, D. Fabiani, Evaluation of dc insulation performance based on space-charge measurements and accelerated life tests, *IEEE Transactions on Dielectrics and Electrical Insulation* 7(3) (2000) 322-328.
- [39] M. Abou-Dakka, A. Bulinski, S. Bamji, Space charge development and breakdown in XLPE under DC field, *IEEE*

- [40] A. Pourrahimi, T. Hoang, D. Liu, L. Pallon, S. Gubanski, R. Olsson, U. Gedde, M. Hedenqvist, Highly efficient interfaces in nanocomposites based on polyethylene and ZnO nano/hierarchical particles: A novel approach toward ultralow electrical conductivity insulations, *Advanced Materials* 28(39) (2016) 8651-8657.
- [41] L. Pallon, A. Hoang, A. Pourrahimi, M. Hedenqvist, F. Nilsson, S. Gubanski, U. Gedde, R. Olsson, The impact of MgO nanoparticle interface in ultra-insulating polyethylene nanocomposites for high voltage DC cables, *Journal of Materials Chemistry A* 4(22) (2016) 8590-8601.
- [42] D. Liu, A. Hoang, A. Pourrahimi, L. Pallon, F. Nilsson, S. Gubanski, R. Olsson, M. Hedenqvist, U. Gedde, Influence of nanoparticle surface coating on electrical conductivity of LDPE/Al₂O₃ nanocomposites for HVDC cable insulations, *IEEE Transactions on Dielectrics and Electrical Insulation* 24(3) (2017) 1396-1404.
- [43] T. Takada, Y. Hayase, Y. Tanaka, Space charge trapping in electrical potential well caused by permanent and induced dipoles for LDPE/MgO nanocomposite, *IEEE Transactions on Dielectrics and Electrical Insulation* 15(1) (2008) 152-160.
- [44] B. Han, X. Wang, Z. Sun, J. Yang, Q. Lei, Space charge suppression induced by deep traps in polyethylene/zeolite nanocomposite, *Applied Physics Letters* 102(1) (2013) 012902.
- [45] C. Li, J. Hu, C. Lin, J. He, The control mechanism of surface traps on surface charge behavior in alumina-filled epoxy composites, *Journal of Physics D: Applied Physics* 49(44) (2016) 445304.
- [46] Q. Li, Q. Wang, Ferroelectric Polymers and Their Energy-Related Applications, *Macromolecular Chemistry and Physics* 217(11) (2016) 1228-1244.
- [47] Q. Li, F. Liu, T. Yang, M.R. Gadinski, G. Zhang, L.-Q. Chen, Q. Wang, Sandwich-structured polymer nanocomposites with high energy density and great charge-discharge efficiency at elevated temperatures, *Proceedings of the National Academy of Sciences* 113(36) (2016) 9995-10000.
- [48] X. Li, Q. Du, J. Kang, D. Tu, Influence of microstructure on space charges of polypropylene, *Journal of Polymer Science Part B: Polymer Physics* 40(4) (2002) 365-374.

- [49] T. Tanaka, M. Kozako, N. Fuse, Y. Ohki, Proposal of a multi-core model for polymer nanocomposite dielectrics, IEEE Transactions on Dielectrics and Electrical Insulation 12(4) (2005) 669-681.
- [50] T. Tanaka, G. Montanari, R. Mulhaupt, Polymer nanocomposites as dielectrics and electrical insulation-perspectives for processing technologies, material characterization and future applications, IEEE Transactions on Dielectrics and Electrical Insulation 11(5) (2004) 763-784.